



Nanosheet-structured K–Co–MoS₂ catalyst for the higher alcohol synthesis from syngas: Synthesis and activation

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ABSTRACT

The nanosheets structured K–Co–MoS₂ catalyst was prepared through a one-step hydrothermal synthesis combined with the wetness impregnation. The fresh catalyst has a high dispersion of Co–Mo–S active phase and no Co₉S₈ is found. The pure H₂ activated catalyst shows a higher intrinsic activity, especially the C₂₊OH selectivity for the higher alcohol synthesis compared to the one activated by 5% H₂/N₂ atmosphere. The reason is attributed to that the pure H₂ activation more effectively suppresses the formation of Co₉S₈ and stabilizes the Co–Mo–S active phase during the reaction due to the formation of SH species.

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1. Introduction

Higher alcohols usually include C₁ to C₆ alcohols, which have good value for use as cleaning fuels, petrochemical feedstocks and gasoline additives for increasing the octane number. The synthesis of higher alcohol from syngas (derived from coal, biomass and natural gas) has attracted wide attention due to the scarcity of energy resources and environmental protection. In the past few decades, several catalyst systems for the synthesis of higher alcohols have been developed, including noble metal Rh-based catalysts, modified methanol synthesis catalysts, modified Fischer–Tropsch (FT) synthesis catalysts, and Mo-based catalysts [1].

Initially, MoS₂-based catalysts were mainly used in hydrodesulfurization in the petroleum industry, which have the advantages of no noble metal component, high resistance to sulfur poisoning and less deactivation by coking [2]. In the 1980s, Dow Chemical Company reported that the alkali metal promoted MoS₂-based catalyst shows high activity and selectivity for the higher alcohol synthesis from syngas [3]. The addition of alkali metal promotes the associative adsorption of CO through a formate mechanism and reduces the hydrogenation activity of the catalyst, thereby increasing significantly the selectivity and STY to alcohols. On the other hand,

the alkali metal promoters can neutralize the surface acidity of catalyst, inhibiting the dehydration reaction of alcohol products [4]. Further research found that the addition of Co, Ni, Rh and Pd promoters is favorable for improving the alcohol production, especially the C₂₊OH selectivity [5].

For a K-modified MoS₂ catalyst, the incorporation of cobalt can result in the substantial structure changes and promote the alcohol chain growth. The incorporated Co atoms mainly exist in two major forms: Co–Mo–S or Co₉S₈ phases, which have very different catalytic properties. The Co–Mo–S phase is suggested to be the active site for CO non-dissociative/dissociative adsorption, non-dissociative CO insertion, carbon chain growth and thus favorable for the alcohol formation. While the Co₉S₈ phase which promotes CO and H₂ dissociative adsorption is involved in the formation of by-product hydrocarbons and inactive for the alcohol synthesis [6–8]. The composition of the two phases strongly depends on the preparation methods [9]. The common methods for the preparation of alkali modified Co–MoS₂ catalysts include impregnation/coprecipitation methods, reverse microemulsion technique and mechano-chemical activation and so on [6,10,11]. Taking the impregnation as an example, the as-prepared Co–MoS₂ catalyst mainly contained the Co–Mo–S phase if the Co/Mo atomic ratio less than 0.4. Higher cobalt content will result in the formation of Co₉S₈ [12].

The activation of catalyst before reaction also has a significant impact on the catalytic performance, because major physical and chemical restructuring seems to take place during the activa-

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tion. The relevant studies on the bimetallic CoCu synthesis alcohol catalyst reveal that CO-activated catalyst shows higher activity and Anderson–Schulz–Flory (ASF) chain lengthening probability than pure H₂- or syngas-activated catalyst, but alcohol selectivity is relatively lower [13]. Detailed characterization results indicate that major restructuring occurs during activation. The CO activation leads to the segregation of Co on the catalyst surface and forms an “onion-like” graphitic carbon shell. So far, few studies have been made on the activation of Mo catalysts. Our previous work reported that for the Mo-based oxide catalyst, the syngas-activated catalyst shows the highest activity for the synthesis of higher alcohols than that of the pure H₂- or CO-activated catalysts [14]. The reason is attributed to that the syngas activation had an appropriate reduction capacity and produced a large amount of Mo^{δ+} ($1 < \delta < 4$) species and multivalent state Co species on the surface of catalyst. Their synergistic effects enhance the cooperativity and equilibrium between the CO dissociation, hydrogenation and CO insertion and thus promotes the formation of higher alcohols. For the sulfide K₂CO₃/MoS₂ catalyst, Sun et al. reported that the pretreatment with H₂, CO or syngas before reaction could effectively shorten the time of induction period and promote the catalytic activity [15]. The reason is suggested to be related to the sulfur losing and K element dispersion on the surface of catalyst. In this paper, we synthesized a nanosheets structured K–Co–MoS₂ by a one-step hydrothermal synthesis combined with the wetness impregnation and investigated the influence of the activation procedure on the catalytic performance. The catalyst structures were characterized by X-ray diffraction (XRD), Transmission electron microscopy (TEM) and X-ray photoelectron spectroscopy (XPS), and the catalytic performance for the higher alcohols synthesis was investigated.

2. Experimental

2.1. Catalyst preparation

The K–Co–MoS₂ catalysts were prepared by a hydrothermal method combined with incipient wetness impregnation. (NH₄)₆Mo₇O₂₄·4H₂O (99.0%), Co(NO₃)₂·6H₂O (98.5%), CS(NH₂)₂ (99.0%) and K₂CO₃ (99.0%) were supplied by Sinopharm Chemical Reagent Co., Ltd and used as received without any pretreatment.

Firstly, the Co–MoS₂ composite was prepared by a hydrothermal method. In a typical procedure, 1.3568 g of (NH₄)₆Mo₇O₂₄·4H₂O was dissolved in 60 mL of deionized water. After complete dissolving, 2.3400 g of CS(NH₂)₂ and 0.8947 g of Co(NO₃)₂·6H₂O were added in turn to the solution under stirring, and formed a red transparent solution. The solution volume was adjusted to 90 mL by adding deionized water. After stirring for 30 min, the mixed solution was transferred to a 150 mL Teflon-lined autoclave. Then the Teflon-lined autoclave was sealed and heated at 473 K for 24 h. After cooling to room temperature, the obtained black precipitate was collected, washed by ethanol and deionized water for several times, and then dried at 323 K for 12 h. The as-prepared Co–MoS₂ was then impregnated with K₂CO₃ aqueous solution. After ultrasonic dispersion for 20 min, the mixture was dried at 338 K for 3 h and calcined in flowing argon at 723 K for 90 min. The atomic ratios of K/Mo and Co/Mo were 0.5 and 0.6, respectively.

2.2. Catalyst characterization

Power X-ray diffraction (XRD) patterns were obtained using a Rigaku D/max-γA rotating-anode diffractometer equipped with a Cu Kα radiation source at 40 kV and 200 mA with a scan range of 10°–70°. Transmission electron microscopy (TEM) measurements were performed using a JEM-2100F field emission transmission

electron microscopy, a JEM-2010 transmission electron microscopy and a H-7650 transmission electron microscopy. X-ray photoelectron spectroscopy (XPS) was collected on a Thermo ESCALAB 250 X-ray photoelectron spectrometer. The binding energy of C 1s (284.8 eV) was used as a reference to correct the binding energy of the catalysts. The Brunauer–Emmett–Teller (BET) surface area was determined by nitrogen adsorption at 77 K using a Micromeritics TriStar II 3020 analyzer.

2.3. Catalytic activity measurement

The catalytic activity was measured in a fixed bed stainless steel reactor with an internal diameter of 8 mm. In each experiment, 0.5 g of catalyst was mixed with quartz sand to obtain a total volume of 2 mL, which was then loaded in the reactor tube. Before reaction, the catalyst was first activated in a flow of activation gas (pure H₂ or 5% H₂/N₂) under 400 °C for 8 h. Then the reactor temperature was lowered to the reaction temperature and syngas (30% CO, 60% H₂ and 10% N₂) was introduced into the reactor. The product gas was cooled down in a trap surrounded by ice-water and separated into gas and liquid phase under reaction pressure. The details of the product analysis were described in our previous work [16]. All activity measurements were performed at 5.0 MPa, 583 K, and Gas Hourly Space Velocity (GHSV) of 4800 h^{−1}. Because the alcohol synthesis required an induction period, all the activity data shown in this study was measured after more than 48 h of reaction.

The intrinsic activity of the catalyst, as expressed by the approximate turnover frequencies (TOFs) for C₂₊OH formation and CO conversion was estimated assuming all surface Mo atoms are active. The accurate TOFs cannot be assessed due to the lack of a suitable probe molecule to titrate the amount of active sites [17]. In this work, the surface amount of Mo atoms was calculated based on the XPS and BET results, as suggested by the literatures [18,19].

The TOFs values for C₂₊OH formation and CO conversion were calculated according to the following equations.

$$\text{TOF}_{\text{C}_{2+\text{OH}}} (\text{s}^{-1} \cdot \text{per surface Mo}) = \frac{\text{mol C}_{2+\text{OH}} (\text{s}^{-1}) \cdot N_A}{\text{No. surface Mo atoms}}$$

$$\text{TOF}_{\text{CO}} (\text{s}^{-1} \cdot \text{per surface Mo}) = \frac{\text{mol reacted CO (s}^{-1}) \cdot N_A}{\text{No. surface Mo atoms}}$$

Where N_A is the Avogadro constant, the No. surface Mo atoms represents the amount of surface Mo atoms, which is determined by the following formula.

$$\begin{aligned} \text{No. surface Mo atoms} &= \text{No. surface total atoms} \\ &\quad \times \text{surface Mo atomic content (\%)} \\ \text{No. surface total atoms} &= \frac{\text{surface area (0.5 g catalyst)}}{\sum_{i=1 \dots n} X_i \times A_i} \end{aligned}$$

Where X_i is the surface atomic content of i atom, which is obtained from the XPS result. A_i is the surface area of i atom, which is estimated according to the ion radius. The BET surface area of catalyst is 0.9363 m²/g.

3. Results and discussion

3.1. XRD results

The XRD patterns of the catalysts activated in different atmospheres and then reacted for 48 h are shown in Fig. 1. For comparison, the XRD pattern of the fresh sample is also presented. The fresh K–Co–MoS₂ catalyst shows very weak diffraction intensities, indicating a low crystallization degree. The peaks at 13.7°, 33.1°, 39.8°, 58.9° are attributed to MoS₂ and those at 27.9°, 36.2°,

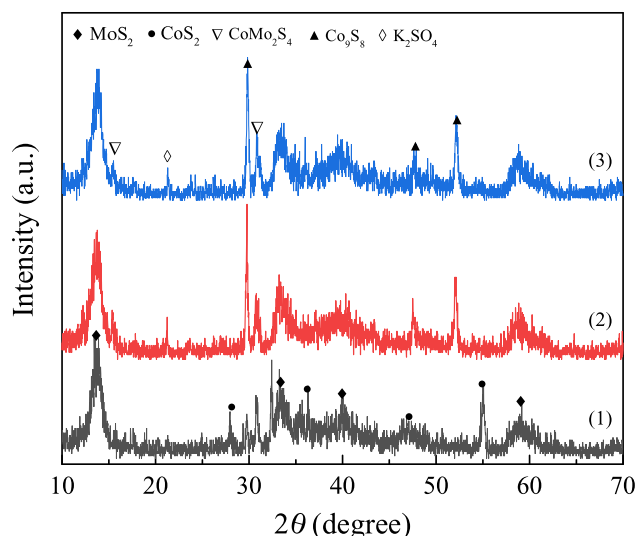


Fig. 1. XRD patterns of (1) the fresh K-Co-MoS₂, (2) the used catalyst pre-activated in 5% H₂/N₂, and (3) the used catalyst pre-activated in pure H₂.

46.7°, 55.0° correspond to CoS₂ species. Besides, the CoMo₂S₄ ($2\theta = 15.4^\circ, 31.1^\circ$) phase is also detected. It is worth noting that no any peaks assigned to Co₉S₈ are found. The results indicate that for the preparation of Co-MoS₂ catalysts, hydrothermal synthesis can more effectively inhibit the formation of Co₉S₈ phase as compared to other methods, such as the impregnation or sulfuration [12]. After activated and reacted, the diffraction peaks assigned to Co₉S₈ were observed ($2\theta = 29.8^\circ, 47.5^\circ, 52.1^\circ$), as shown in Fig. 1. The different activation conditions have different impacts on the formation of Co₉S₈. By comparison, the sample activated in pure H₂ shows weaker diffraction intensities of Co₉S₈ than that of the sample activated in 5% H₂/N₂, indicating that the pure H₂ activation inhibits effectively the formation of Co₉S₈ during the reaction.

3.2. TEM results

The morphology and structure of the as-prepared MoS₂, Co-MoS₂ and K-Co-MoS₂ catalyst were characterized by TEM technique. As shown in Fig. 2(a), the MoS₂ exists as thin 2D nanosheets with the lateral size more than 100 nm. Moreover, the MoS₂ nanosheets are interconnected with each other due to high surface energy of 2D materials, forming the 3D petal-like structure and indicating that the few-layer nanosheets are flexible and foldable. After the incorporation of Co and K, as shown in Fig. 2(b and c), the obtained Co-MoS₂ and K-Co-MoS₂ particles remain the layered nanosheets structure and no aggregation occur, indicating that the incorporated Co and K are well dispersed and anchored on the MoS₂ nanosheets. Fig. 2(d) shows the HRTEM image of the K-Co-MoS₂ nanosheets, in which the characteristic interlayer distance of 0.63 nm for 2H-MoS₂ is detected, corresponding to the (002) plane of MoS₂. Furthermore, the nanosheets organized MoS₂ shows exposed edge sites and abundant defects, which are regarded as the active center for CO hydrogenation [20,21]. Another group of lattice fringes with an interplanar spacing of 0.25 nm is assigned to the (210) plane of CoS₂. The HRTEM image clearly confirms that the Co atoms are located at the edges of MoS₂ nanosheets and form a tight binding between each other, displaying the advantage of hydrothermal synthesis. The used catalysts are also investigated by TEM technique. As shown in the Fig. 3, the catalysts activated in different atmospheres show different degrees of aggregation after reaction. The sample activated by 5% H₂/N₂ aggregates obviously and forms some large size, irregular particles after reaction. In contrast, no significant aggregation is observed for the catalyst particles activated by pure H₂, suggesting that the high pure hydrogen can inhibit the aggregation of catalyst particles during the reaction.

3.3. XPS results

The surface chemical composition of the fresh and used K-Co-MoS₂ catalysts is analyzed by XPS technique. Fig. 4 shows the XPS spectra of Mo 3d for the K-Co-MoS₂ catalysts. For the fresh

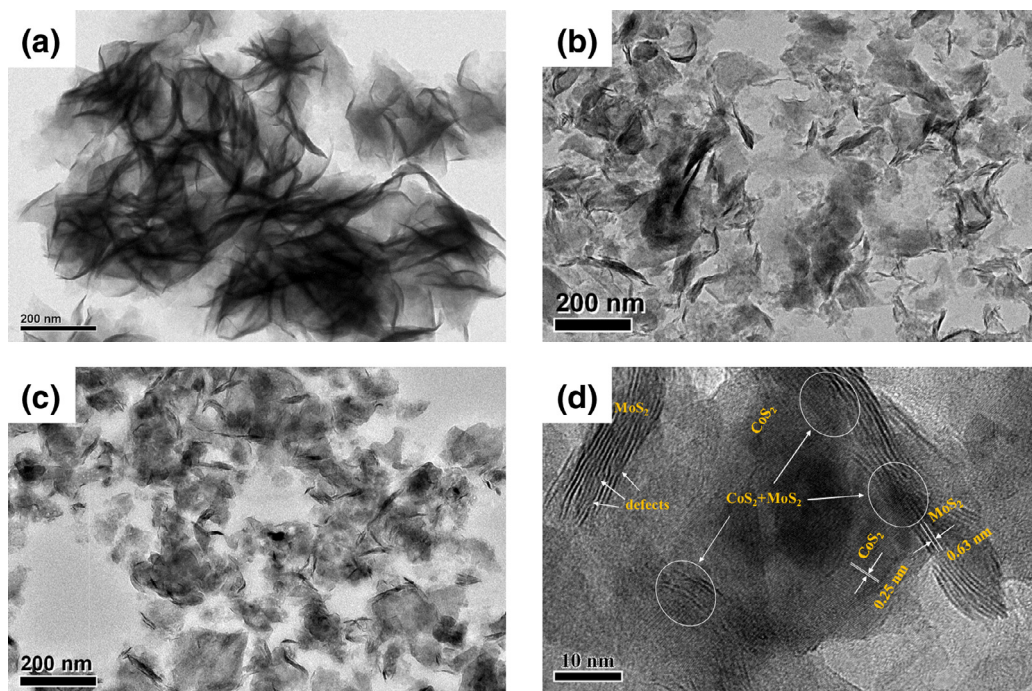


Fig. 2. TEM images of (a) the MoS₂, (b) the Co-MoS₂, and (c) the fresh K-Co-MoS₂; (d) the HR-TEM image of the fresh K-Co-MoS₂.

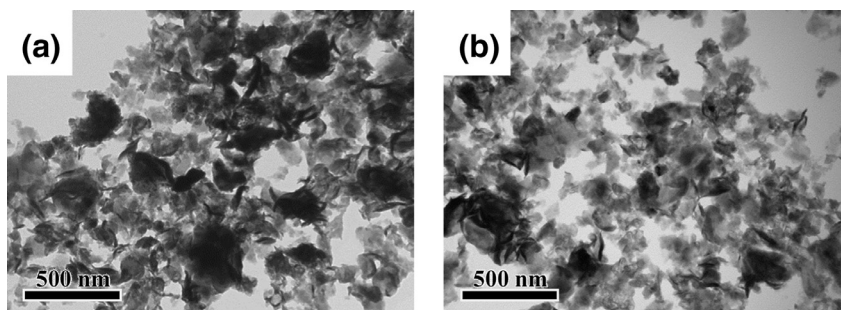


Fig. 3. TEM images of (a) the used catalyst pre-activated in 5% H_2/N_2 , and (b) the used catalyst pre-activated in pure H_2 .

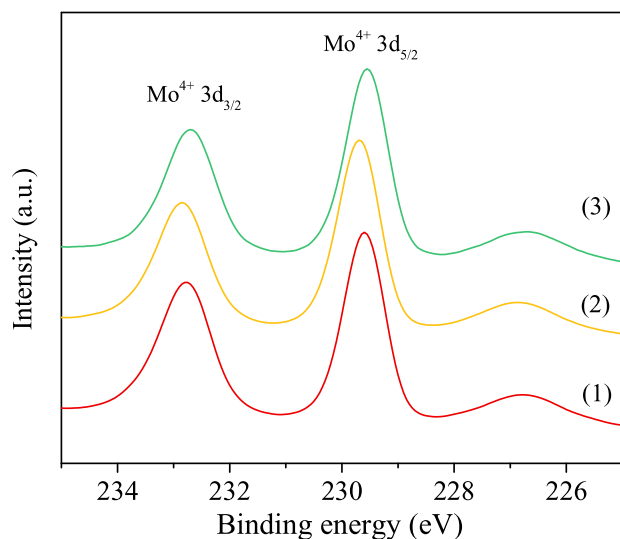


Fig. 4. Mo 3d XPS spectra of (1) the fresh K-Co-MoS₂, (2) the used catalyst pre-activated in 5% H_2/N_2 , and (3) the used catalyst pre-activated in pure H_2 .

K-Co-MoS₂ catalyst, the characteristic binding energies of Mo⁴⁺ 3d_{5/2} (229.5 eV) and Mo⁴⁺ 3d_{3/2} (232.8 eV) are observed. After reaction, the Mo valence state of K-Co-MoS₂ catalysts remains unchanged. It is noted that the binding energy of Mo⁴⁺ of the catalyst activated by pure H_2 shifted obviously to lower levels compared to that of the catalyst activated in 5% H_2/N_2 at same temperature. Similar result is also observed for the S 2p spectra as shown in Fig. 5. The reason can be attributed to that pure H_2 activation promotes the electron transfer from Co to increase the electron cloud density, intensifies the electromagnetic shield for the inner shell electron due to the formation of SH groups, and thus decreases the electron binding energy.

The XPS spectra of Co 2p for the fresh and used K-Co-MoS₂ catalysts are shown in Fig. 6. From the Co 2p core-level spectra, the binding energies of 779.3 eV and 794.2 eV are assigned to Co 2p_{3/2} and Co 2p_{1/2}, respectively, confirming the existence of CoS₂ species [22,23]. Furthermore, the difference of binding energy between Co 2p_{3/2} and Mo 3d_{5/2} with 549.8 eV is relatively close to the value of 550.0 eV reported in the literature with the cobalt atoms located in a MoS₂ phase [24,25], which is regarded as an evidence for the formation of Co-Mo-S phase. Based on the fact and combined with TEM results, it is suggested that Co atoms in CoS₂ are located on the edges of MoS₂ platelets, forming the Co-Mo-S phase. The peaks at 781.8 eV and 798.2 eV and two satellite peaks at 784.1 eV and 801.5 eV are attributed to the Co²⁺ in the Co₉S₈, in good agreement with the previous reports [26]. For the fresh K-Co-MoS₂ catalyst, the Co 2p peak assigned to CoS₂ is strong while that of Co₉S₈ is very weak, indicating that the Co atoms in fresh

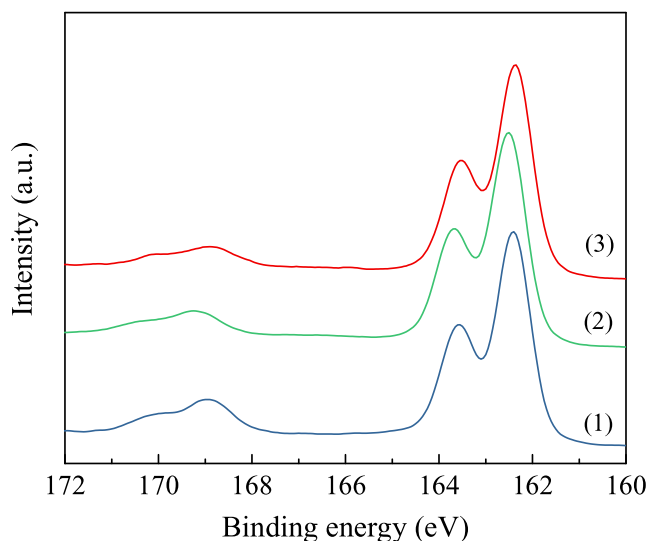


Fig. 5. S 2p XPS spectra of (1) the fresh K-Co-MoS₂, (2) the used catalyst pre-activated in 5% H_2/N_2 , and (3) the used catalyst pre-activated in pure H_2 .

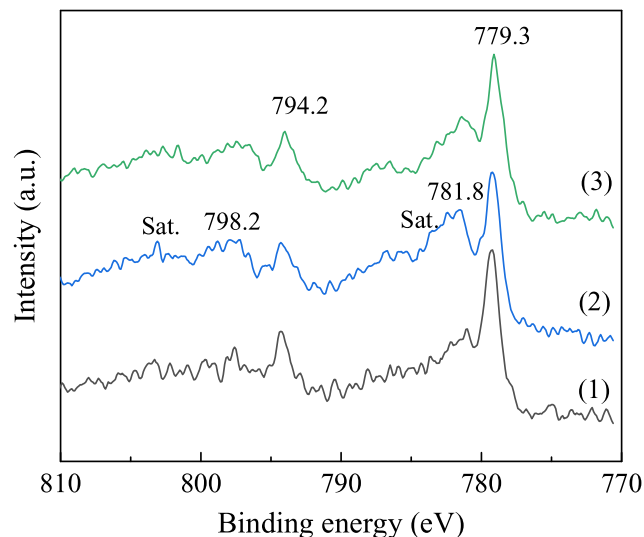
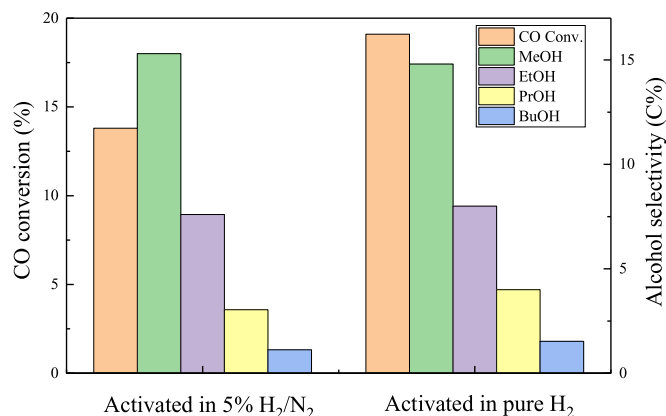


Fig. 6. Co 2p XPS spectra of (1) the fresh K-Co-MoS₂, (2) the used catalyst pre-activated in 5% H_2/N_2 , and (3) the used catalyst pre-activated in pure H_2 .

catalyst are mainly in the form of Co-Mo-S phase. The different activation conditions have different impacts on the existing state of cobalt. After activated by 5% H_2 , the Co 2p peaks corresponding to Co₉S₈ increase significantly after reaction, revealing that a large amount of Co atoms are separated from the Co-Mo-S to form the

Table 1. TOFs values of the catalysts activated in different gases.

Activation gas	CO conv. (C%)	C ₂₊ OH/MeOH (C%)	TOF _{CO} ^a (s ⁻¹)	TOF _{C₂₊OH} ^b (s ⁻¹)
5% H ₂	13.87	0.76	0.27	0.031
Pure H ₂	19.01	0.92	0.33	0.044

^a Based on the CO conversion.^b Based on the C₂₊OH formation.**Fig. 7.** CO conversion and alcohol products distribution of the catalysts activated in different gases.

Co₉S₈. In contrast, the pure H₂-activated catalyst shows relatively weaker Co 2p peaks of Co₉S₈, indicated that the pure H₂ activation can more effectively inhibit the formation of Co₉S₈ during the reaction, in accordance with the XRD and TEM results.

3.4. Catalytic performance

The catalytic performance for higher alcohol synthesis over the K-Co-MoS₂ catalysts was tested and the activity data have been recorded after an induction period of 48 h. As shown in Fig. 7, the catalyst activated in pure H₂ shows the higher activity toward the higher CO conversion and alcohol selectivity compared to the sample activated in 5% H₂. Furthermore, the selectivity to C₂₊OH increases significantly. The intrinsic activity of the catalyst, as expressed by the approximate turnover frequencies (TOFs) for C₂₊OH formation and CO conversion was evaluated assuming all surface Mo atoms are active (Table 1). The accurate TOFs cannot be assessed due to the lack of a suitable probe molecule to titrate the amount of active sites [17]. In this work, the surface amount of Mo atoms was calculated based on the XPS and BET results, as suggested by the literatures [18,19]. First, the surface atomic concentration was obtained through the XPS analysis. Then combined with the BET result and ion radius data, the amount of surface Mo atoms was estimated and used for calculating the TOFs per surface Mo. From the Table 1, the catalyst activated in pure H₂ has the higher intrinsic activity. The TOFs values for CO conversion and C₂₊OH formation are 0.33 s⁻¹ and 0.04 s⁻¹, respectively, higher than that of the Rh-based catalyst [27,28], which is the most selective catalyst for the C₂₊OH production (mainly ethanol).

For the MoS₂-based catalysts, cobalt is an effective promoter for improving the C₂₊OH selectivity and STY of total alcohol. The incorporated Co atoms mainly exist in two major forms: Co-Mo-S or Co₉S₈ phases. Systematic researches on preparation-structure-activity relationship indicate that the two phases have very different catalytic properties [29,30]. Co₉S₈ contributes little to the alcohol production, and it is stable under typical reaction conditions. The Co-Mo-S phase is generally regarded as the active phase for the formation of C₂₊ alcohols, because those active H-species could be easily adsorbed on the Co-Mo-S sites via the hydrogen spillover and thus increasing the rate of hydrogenation reaction [31]. The

generally accepted model of Co-Mo-S phase proposed by Topsøe and Topsøe [32] consists of MoS₂ particles with Co atoms located at the edges of the MoS₂ slabs. The sulfur atom is shared between Co and Mo and the electron transfer from Co to Mo involves an electron transfer from Co-S bond to a Mo-S bond [33].

The content of the two phases (Co-Mo-S and Co₉S₈) strongly depends on the catalyst preparation method and the Co concentration [8]. For the presented catalyst synthesized via the hydrothermal method, the characterization results clearly demonstrate that the Co atoms are mainly located at the edge of MoS₂ nanosheets, forming the Co-Mo-S phase. No separated Co₉S₈ phase is found even if the Co/Mo atomic ratio is up to 0.6, showing an obvious advantage in comparison to the conventional impregnation method [12]. The structural feature leads to a high dispersion of active sites on the catalyst surface, thus showing a high intrinsic activity for higher alcohol synthesis.

The syngas-exposure reaction environment easily results in the separation of the Co atoms from the edge of MoS₂ to form the Co₉S₈, and therefore reduces the catalyst activity, especially the C₂₊OH selectivity [34]. The catalyst activation before reaction can inhibit effectively the formation of Co₉S₈. For the presented K-Co-MoS₂ catalyst, the characterization results indicate that the pure H₂ activation can suppress the separation of Co atoms maximally and remains most of the Co atoms in the form of Co-Mo-S phase compared to 5% H₂/N₂ activation gas. The reason is attributed that in pure hydrogen atmosphere, the sulfur on the Co-Mo-S edge can be more easily hydrogenated to form the SH groups. The formation of SH groups promotes the electron transfer from Co to Mo through sulfur atoms and enhances the anchoring of Co to MoS₂, therefore stabilizing the Co-Mo-S phase [35]. Furthermore, the formation of SH groups also leads to a low sulfur-covered Co-Mo-S edge and a significant increase in coordinatively unsaturated site (CUS). The abundant CUS provides more Lewis acidity site, thus enhancing CO adsorption and improving the catalyst activity [36]. Consequently, the catalyst activated in pure H₂ shows higher activity, especially the C₂₊OH selectivity toward the higher alcohols synthesis. The work provides some new insights for the preparation and activation mechanism of Mo-based catalysts for the higher alcohol synthesis.

4. Conclusions

We synthesized a nanosheets structured K-Co-MoS₂ by a one-step hydrothermal synthesis combined with the wetness impregnation. The fresh catalyst has a high dispersion of Co-Mo-S active phase and No Co₉S₈ is found. The pure H₂ activation before reaction suppresses maximally the formation of Co₉S₈ and stabilizes the Co-Mo-S active phase during the reaction, thus showing a high intrinsic activity for the higher alcohol synthesis. It is suggested that the pure H₂ atmosphere is favorable to hydrogenate the sulfur on the Co-Mo-S to form SH groups, which promotes the electron transfer from Co to Mo and thus stabilizes the Co-Mo-S active phase.

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